WORKED SOLUTION

STEP II 1998, Question 12

Probability and Statistics • Bayes' Theorem

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Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/98-s2-q12

This question has a full guided treatment in *Beyond STEP*, Section 12.2, *The Base-Rate Fallacy*: https://beyondstep.keithmcnulty.org/ch12/base-rate.html

The key idea

This is Bayes' theorem, twice. The first part is the classic "rare disease" surprise: when the disease is rare, most positives can be false positives even for a fairly accurate test. For the second part, the trick is to see that the people who take STEP are not a random sample of the population. They are the AL-positive group, so the answer to the first part becomes the *prior* for the second test.

Solution

Let D be the event that a person has mathematitis, A the event that test AL is positive, and B the event that test STEP is positive. We are given

$$\mathrm{P}(D) = \tfrac{1}{100}, \qquad \mathrm{P}(A \mid D) = \tfrac{9}{10}, \qquad \mathrm{P}(A \mid D') = \tfrac{1}{11}.$$

Frankie's probability

By the law of total probability,

$$\begin{aligned}\mathrm{P}(A) &= \mathrm{P}(A \mid D)\,\mathrm{P}(D) + \mathrm{P}(A \mid D')\,\mathrm{P}(D') \\ &= \frac{9}{10}\cdot\frac{1}{100} + \frac{1}{11}\cdot\frac{99}{100} = \frac{9}{1000} + \frac{9}{100} = \frac{99}{1000}.\end{aligned}$$

Then by Bayes' theorem,

$$\mathrm{P}(D \mid A) = \frac{\mathrm{P}(A \mid D)\,\mathrm{P}(D)}{\mathrm{P}(A)} = \frac{9/1000}{99/1000} = \frac{1}{11}.$$

So even after a positive AL result, Frankie is a sufferer with probability only $\frac{1}{11}$. In a group of 1000 people, about 9 sufferers test positive but so do about 90 of the 990

non-sufferers, and the false positives swamp the true ones.

Finding p

Now restrict attention to the people for whom AL was positive. Within this group,

$$\mathrm{P}(D \mid A) = \tfrac{1}{11}, \qquad \mathrm{P}(D' \mid A) = \tfrac{10}{11}.$$

We read the given information about STEP as $\mathrm{P}(B \mid D \cap A) = 0.9$ and $\mathrm{P}(B \mid D' \cap A) = p$: STEP's behaviour depends only on whether the person is a sufferer, not on what AL said.

Ask yourself Why don't we start again from $\mathrm{P}(D) = \frac{1}{100}$?

Because STEP is only given to people who were AL-positive, and in that group the proportion of sufferers is no longer 1%; it is $\frac{1}{11}$, which is exactly what the first part found. The posterior from the first test becomes the prior for the second.

Within the AL-positive group,

$$\mathrm{P}(D \cap B \mid A) = \frac{1}{11} \cdot \frac{9}{10} = \frac{9}{110},$$
$$\mathrm{P}(D' \cap B \mid A) = \frac{10}{11} \cdot p = \frac{10p}{11}.$$

We are told that half of those who are positive on both tests are sufferers, that is $\mathrm{P}(D \mid A \cap B) = \frac{1}{2}$. This means the two probabilities above are equal:

$$\frac{9}{110} = \frac{10p}{11} \quad \Longrightarrow \quad p = \frac{9}{100} = 0.09.$$

Checks

- Odds form: the prior odds of disease are $1 : 99$. AL multiplies them by $\frac{0.9}{1/11} = 9.9$, giving $9.9 : 99 = 1 : 10$, which matches $\frac{1}{11}$. To reach odds $1 : 1$, STEP must multiply by 10, so $\frac{0.9}{p} = 10$ and $p = 0.09$.
- A simulation of two million people through both tests (with $p = 0.09$) gave 0.0909 for the proportion of sufferers among AL-positives and 0.5003 among those positive on both tests.

Common slips

- Using $\mathrm{P}(D) = 0.01$ as the prior for STEP, which gives $p = 0.0009$, a hundred times too small.
- Confusing $\mathrm{P}(A \mid D) = 0.9$ with $\mathrm{P}(D \mid A)$. The whole point of the first part is that they are very different.

Beyond STEP

Writing Bayes' theorem in odds form, with posterior odds equal to prior odds times the likelihood ratio, makes sequential testing almost mechanical: each independent piece of evidence just multiplies the odds by its own factor. This is the basis of Bayesian inference and of how medical screening programmes are actually designed. The assumption we made, that the tests are conditionally independent given disease status, is the same one behind "naive Bayes" classifiers. Blitzstein and Hwang, *Introduction to Probability*, has an excellent treatment of conditional probability and Bayes' rule with many examples like this one.

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